Practical guide to machine learning potentials

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Machine learning potentials (MLPs) in a nutshell

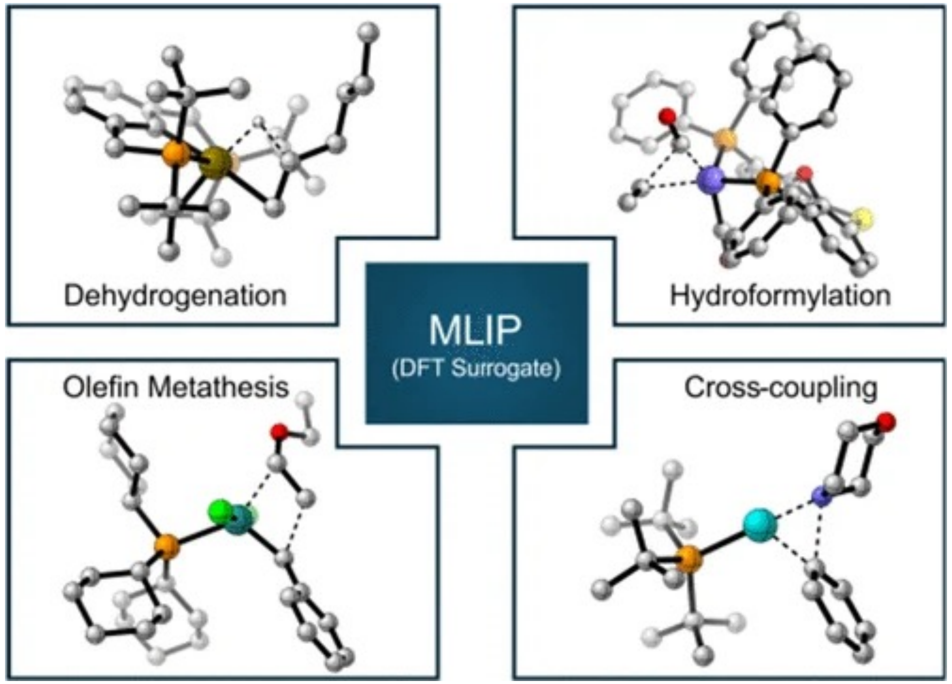
- ML models that predict the energies, forces etc. from
  - 3D atomic coordinates
  - Spin state, total charge
- Can replace slow DFT simulations for many applications

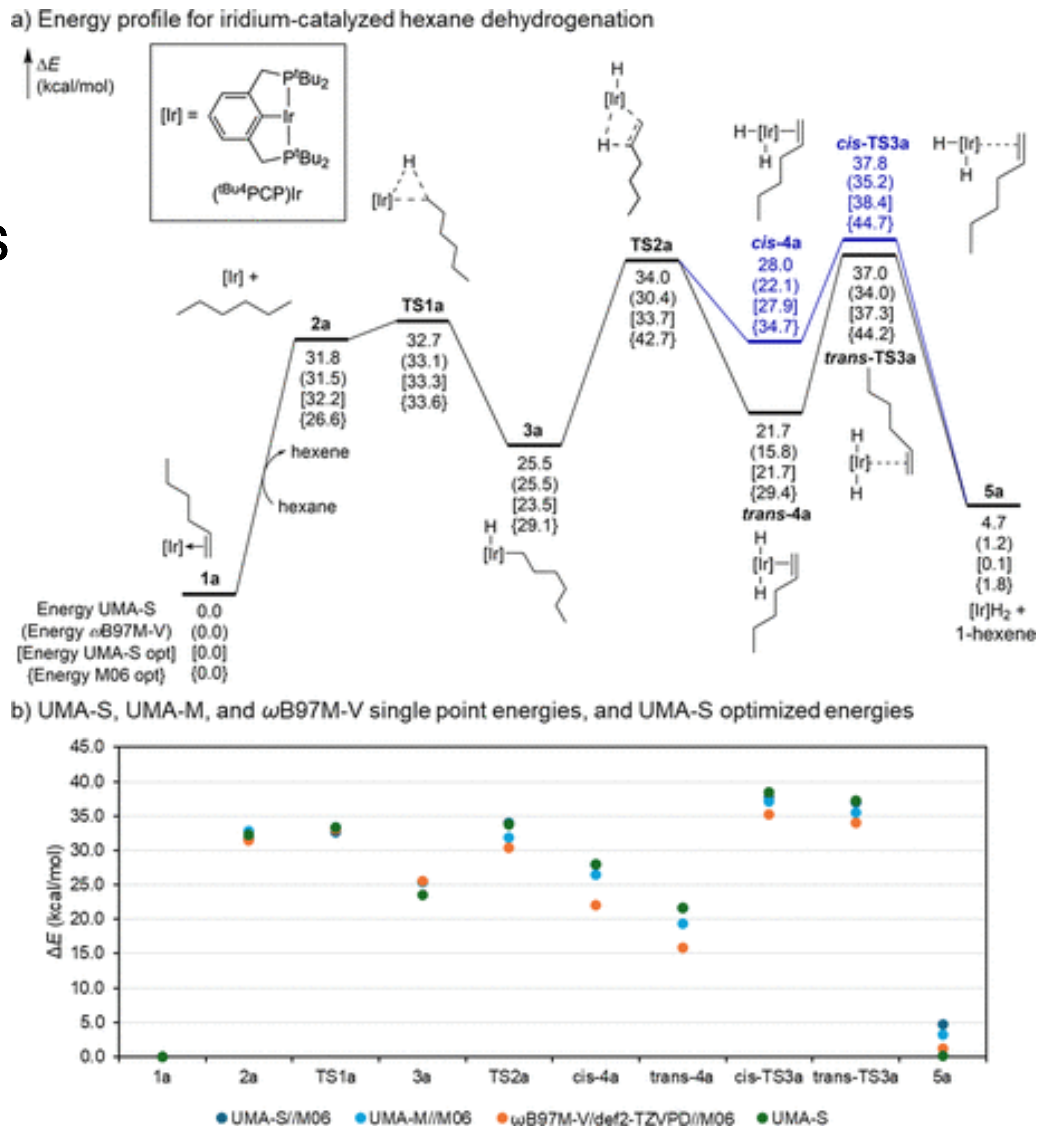
Increasing computational cost →

Force fields	Machine learning potentials	Semi-empirical methods	Density functional theory	Ab initio
GFN-FF	ANI-1ccx	GFN2-xTB	B3LYP-D3	CCSD(T)
MMFF94	ANI-2x	PM6	ωB97XD	ADC(2)
UFF	AIQM1	AM1	r2SCAN-3c	CASSCF
…	MACE-OFF23	…	ωB97X-3c	CASPT2
	…		…	…

Applied to reactions

- Orders of magnitude faster simulations





https://doi.org/10.1021/acscatal.5c09093

Different generations of MLPs

- Pre-universal MLPs: only worked for specific molecules/systems
- First-generation universal MLPs: only specific combinations of elements (*e.g.*, C, H, O, N), structures (equilibrium) etc.
- Second-generation universal MLPs: Worked well across elements and structures
- Third-generation universal MLPs: Introduce more “physics” such as spins, charges, external electric fields etc.

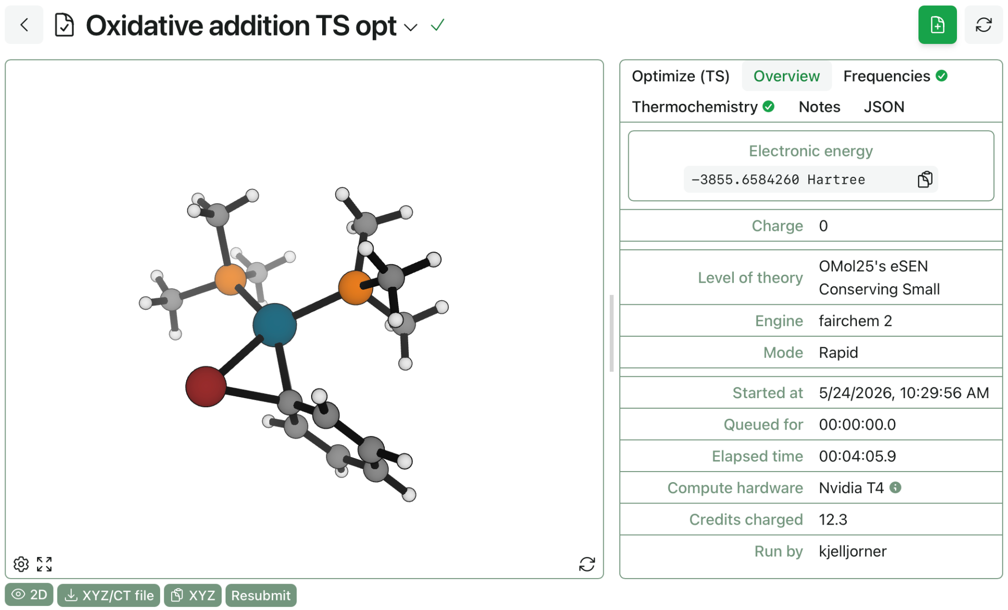
Pre-universal MLPs	**1st universal MLPs**	**2nd universal MLPs**	**3rd universal MLPs**
BPNN	MACE-MP0	UMA	MACE-Polar
	ANI-2x		

How to run MLPs (in order of ease of use)

1. Online platforms such as Rowan Scientific (at cost)
2. Through ORCA's ExtOpt keyword
3. Via Python code using ASE (or other packages)

- You need a GPU for best performance

What we will do today



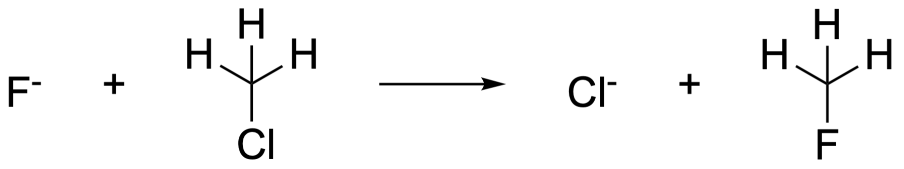
Outline of tutorial

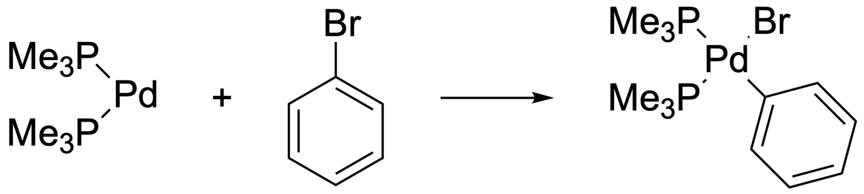
Do the notebooks in the following order (might not be time for all)

1. test_environment.ipynb: Testing that everything works
2. reactive_mlps.ipynb: MLPs for reaction modeling
3. descriptors.ipynb: Extracting descriptors from MLPs

Machine learning potentials for reaction optimization

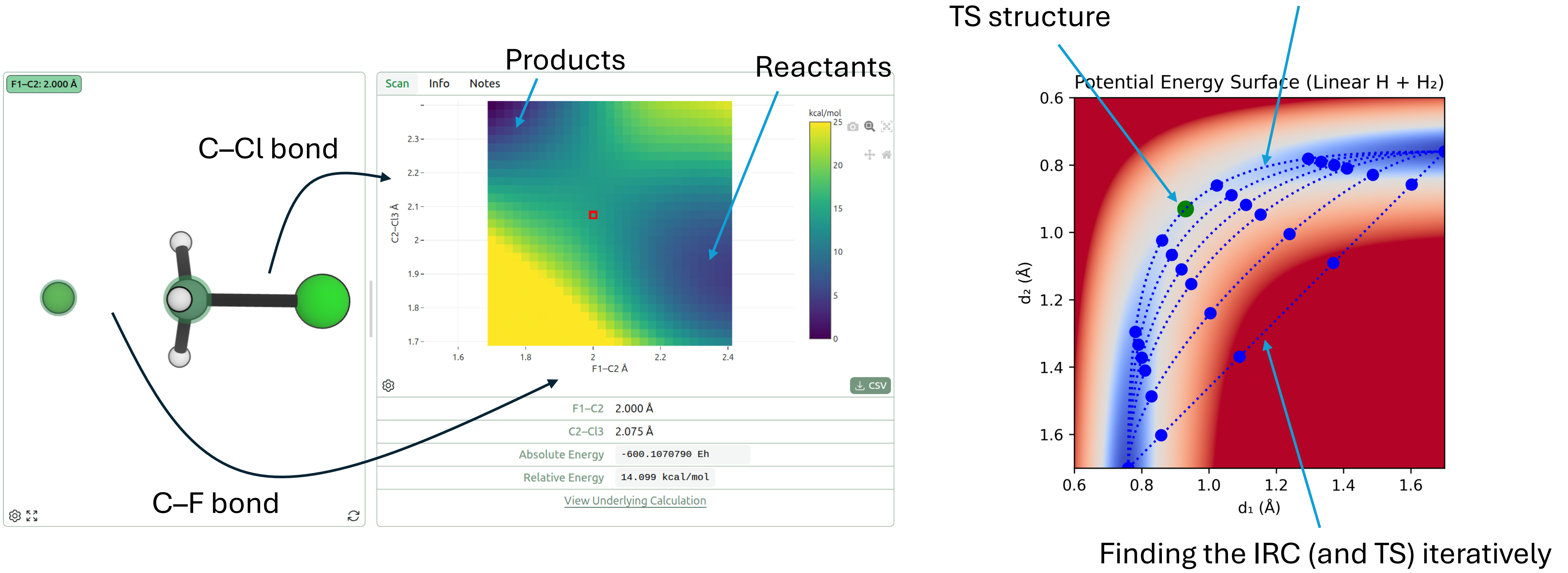
- How good are MLPs for transition states and reaction paths?
- Two reactions that are potentially challenging:
  - S_N2 reaction (small charged nucleophiles/leaving groups)
  - Oxidative addition of bromobenzene to Pd (metal atom, heavy elements)





The potential energy surface

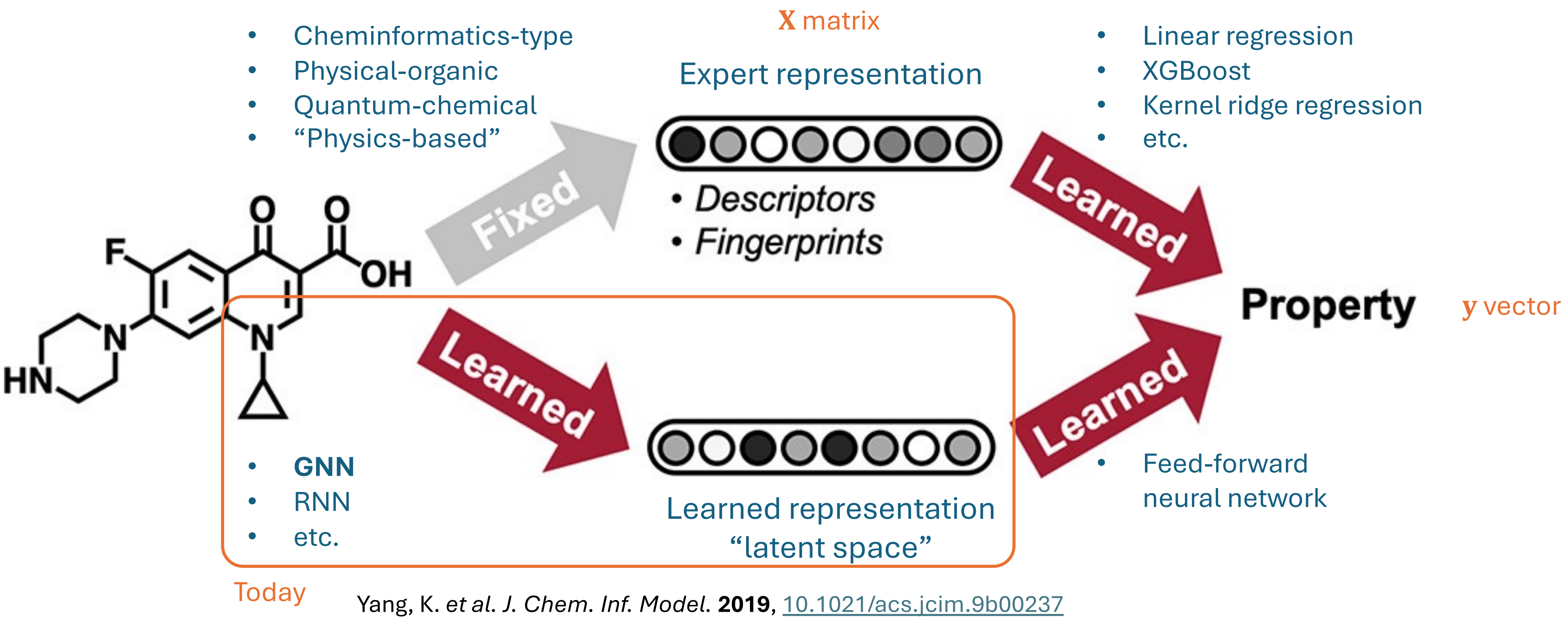
- Describes reaction in terms of coordinates of interest



https://rowansci.com/blog/guessing-transition-states

Descriptors from machine learning potentials

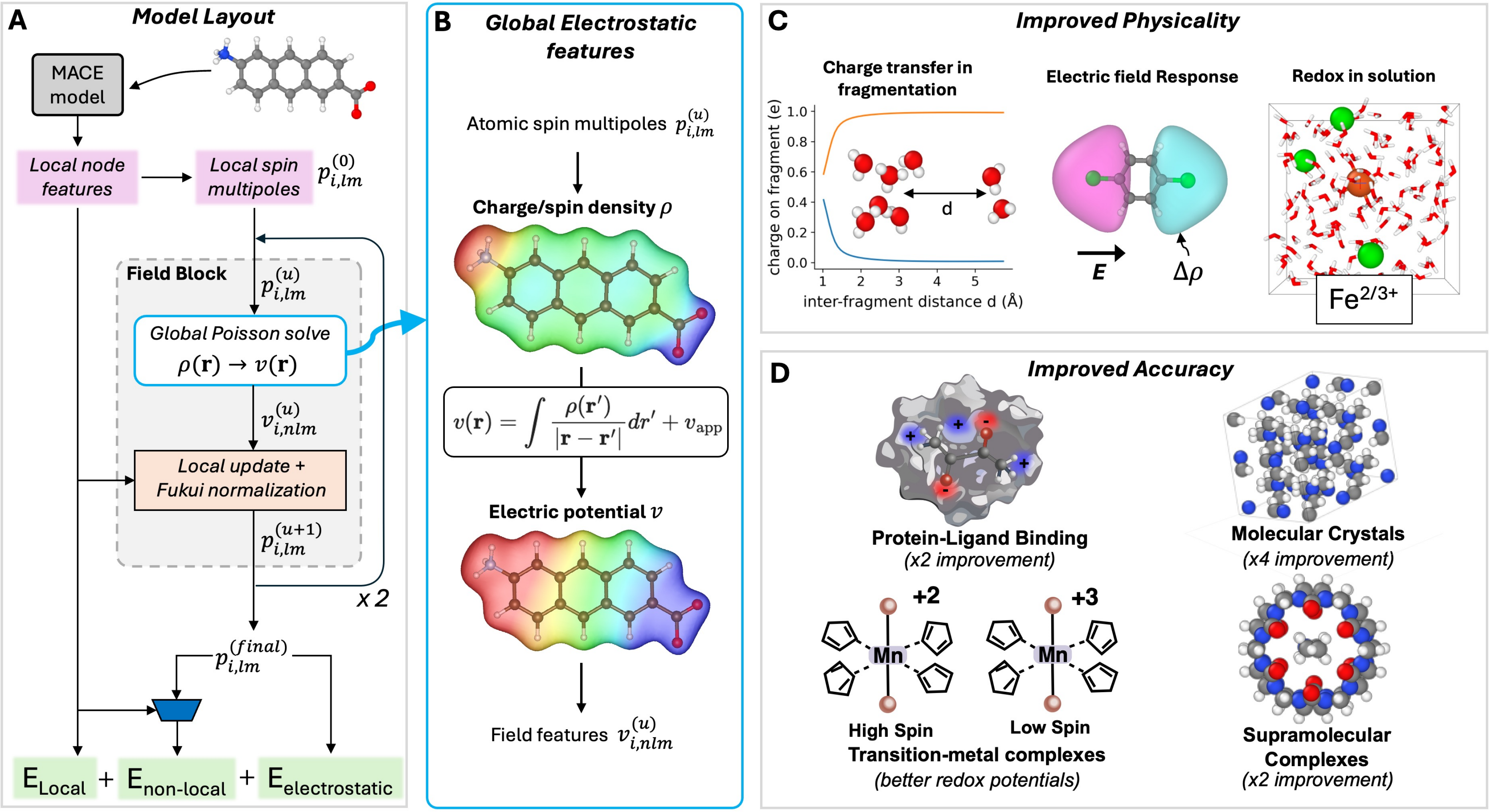
- Deep learning models such as MLPs learn an internal vector representation of the data ⇒ “descriptors”
- These descriptors can be used for other targets than energy



Yang, K. *et al.* *J. Chem. Inf. Model.* **2019**, 10.1021/acs.jcim.9b00237 http://arxiv.org/abs/2602.19411v1

"Third-generation" MLPs

- Allow us to extract atomic spins, charges, dipole moments etc



Questions

Time to work!